

Introduction to MATLAB

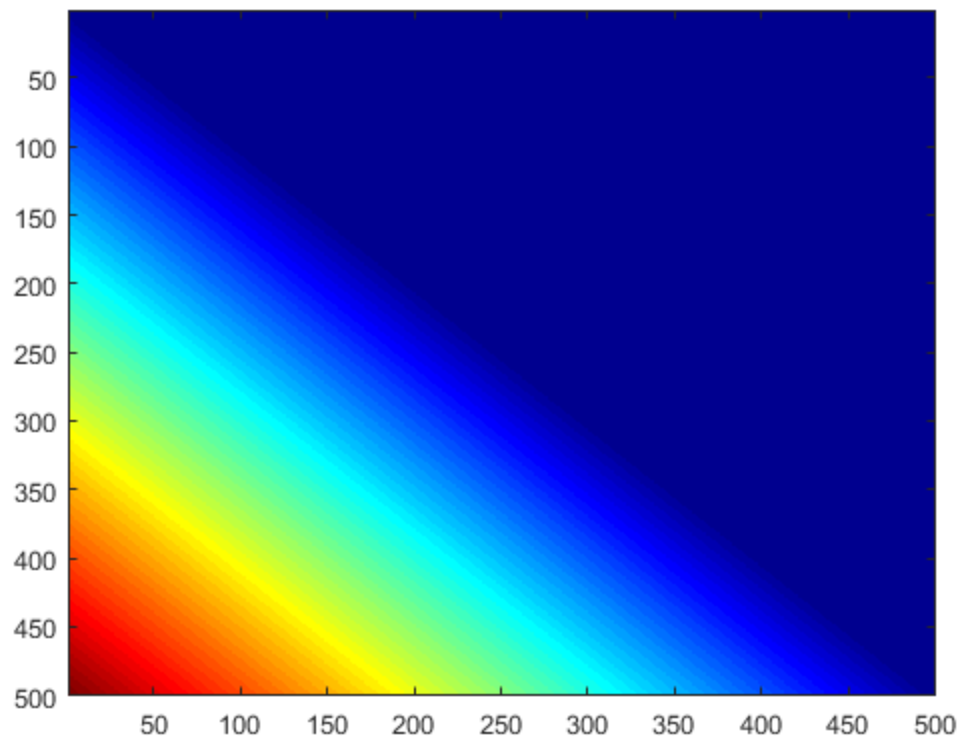
The MATLAB_DEMO.m script gives examples of many basic MATLAB operations. Take some time to familiarize yourself with simple math operations, creation and indexing of vectors and matrices, and logical operations in MATLAB. The second section includes an example of if, for, and while statements. Check that you understand the form of these statements in MATLAB.

Problems:

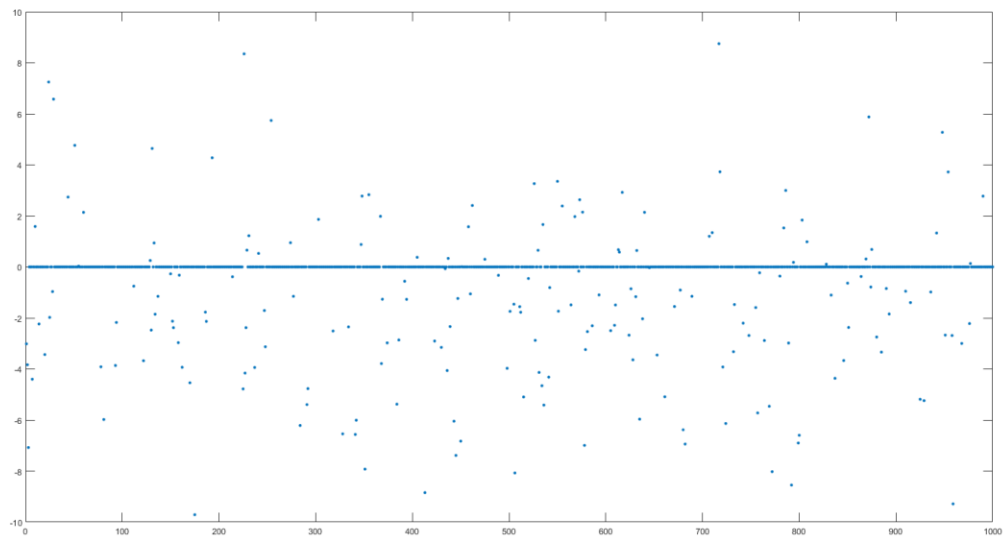
1. Construct the matrix below.

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & \dots & 0 \\ 2 & 1 & 0 & 0 & 0 & \dots & 0 \\ 3 & 2 & 1 & 0 & 0 & \dots & \\ 4 & 3 & 2 & 1 & 0 & & \\ \vdots & \vdots & \vdots & & & \ddots & \ddots \\ 499 & 498 & 497 & \dots & & 1 & 0 \\ 500 & 499 & 498 & \dots & & 2 & 1 \end{bmatrix}$$

, and make a MATLAB plot of it, like the one



2. Create a 1000 element vector. Randomly choose 200 entries of this vector and assign each a random value, drawn from a distribution with standard deviation 3.0 and mean -2.0. The function nnz() returns the number of non-zero elements in its argument. Plot your result, as below.



3. Randomly assign x and y values for four random points. Calculate the coefficients of the cubic function that fits all four points. Plot both the points and the cubic function you calculated, as below.

